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STABLECOIN OPERATIONAL RISK · SCENARIO MODELING · TOKEN-ECONOMIC MODELING

· Digital assets since 2016 (10 years) · 74-vector risk taxonomy · 64 testable controls · 624-scenario simulation battery
· Thorchain $200M insolvency modeled 13 months ahead · 8 SSRN papers · 14 federal comment letters
· 16 USPTO patent applications across 8 families (Canonical-State Architecture for multi-agent AI research)

SUMMARY

Ten years in digital assets: independent protocol research from Jul 2016, institutional practice from Dec 2019. Operational and token-economic risk modeler who turns scenario simulation and coalition analysis into pre-failure risk calls and no-exposure positions, from protocol insolvency to governance capture. Founder of Tokenization Systems; first hire at Borderless Capital, an RIA.

Focus: operational risk, token-economic modeling, scenario modeling, stablecoin and tokenization risk · Monte Carlo, difference-in-differences, PID controller design, supply-equilibrium and coalition-power analysis · OCC, FDIC, FinCEN, Treasury GENIUS, MiCA.

EXPERIENCE

Founder · Tokenization Systems *Feb 2026 to Present*

- Authored an eight-paper SSRN-live research program covering the Control Layer Intensity Index (CLII) for gateway concentration, the Minimum Viable Equivalence Pack (MVEP) for institutional tokenization diligence, programmable monetary-policy design (Adaptive Tokenomics), governance concentration, and operational risk in token economies. Three additional papers in development (B5 Bankruptcy-Priority Misaligned Incentives, A7 Control Layer Theory, SVB Reserve Concentration); methodology paper in preparation.
- Contributed fourteen comment letters across a federal regulatory program (OCC bank stablecoin engagement RIN 1557-AF41 plus supplemental control-layer extension; Treasury GENIUS Act state similarity; FDIC payment-stablecoin reserve standards RIN 3064-AG19 and stablecoin-subsidiary issuance approvals RIN 3064-AG20; FinCEN AML/CFT FINCEN-2026-0034; FinCEN-OFAC PPSI sanctions FINCEN-2026-0100; joint banking AML/CFT 91 FR 18304); Routing the Dollar prepared for the Federal Reserve Fifth Conference on the Dollar's International Roles (Jun 2026); Governance Concentration accepted for publication at Frontiers in Blockchain.
- Active dialogues across U.S. national-security research, major securities exchanges, central securities depositories, global payments messaging infrastructure, tier-1 bank tokenization platforms, and SEC-registered transfer agents on stablecoin, deposit-token, tokenized-deposit, tokenized money-market-fund, tokenized-commodity, and tokenized-equity deployment.

Senior Investment Analyst, Digital Assets · Borderless Capital (RIA; first hire) *Dec 2019 to Feb 2026*

Risk Detection & Loss Prevention

- Predicted Thorchain's **$200M insolvency 13 months in advance**, leading to a no-exposure posture before the predicted failure mode materialized.
- Prevented hostile governance capture on a top-25 blockchain via voting-path simulation under coalition-attack scenarios; surfaced the vulnerability and mitigation route before exploitation.
- Identified ~15% misreporting of circulating supply in a top-25 Layer 1 protocol by independently modeling its tokenomics (emission schedule, vesting, burn dynamics); the discrepancy had propagated to CoinMarketCap and CoinGecko, and was corrected after partnering with the protocol's foundation engineers.

Controls & Risk Frameworks

- Built operational-risk taxonomies for tokenized systems: 74 risk vectors mapped to 64 testable controls and 31 program objectives across a 624-scenario simulation battery (published as Operational Risk in Token Economies); Lido V3 staking-vault production case study sets operator collateral by risk-score alignment.

Diligence & Advisory

- Translated regulatory change (GENIUS, CLARITY, Singapore MAS, MiCA, state money-transmitter and trust-charter regimes) into product, reserve, routing, and compliance decisions across U.S., Singapore, Cayman / BVI / Bermuda, and EU; coordinated cross-functional partner work spanning Product / Engineering, Risk (operational-risk taxonomy), and Legal.
- Built quantitative token-economic models forecasting emissions, fee capture, subsidy-to-revenue trajectories, and governance concentration under wide scenario ranges; supply-equilibrium and slashing dynamics supporting Layer 1 mechanism-design advisory.

Blockchain Research Analyst (Independent) · Self-Employed *Jul 2016 to Present*

- Independent digital-asset research since Jul 2016, three years before joining Borderless Capital: reviewed 250+ protocol whitepapers, diligenced 1,000+ project and token-offering pitches, consulted 150+ projects on protocol design and tokenomics, and invested in 50+ token offerings.
- Hands-on validation across 30+ blockchains under live conditions: liquidity, fees, settlement finality, bridging, governance execution, and custody and security trade-offs. 30+ conferences and bank summits across the U.S., Europe, Asia, and Dubai.

Analyst Intern · Greenway Solutions (fraud-control consulting; client: USAA) *May to Sep 2016*

- Benchmarked fraud controls and the security-versus-friction tradeoff across online, mobile, and call-center channels at nine leading U.S. banks: authentication and password-reset controls, account-opening data requirements, and funds-availability and clearing timing; built the scoring framework and delivered the executive summary.

SELECTED RESEARCH

- **Operational Risk in Token Economies.** Operational-risk taxonomy (74 risk vectors, 64 testable controls, 31 program objectives) mapped to a 624-scenario battery; Lido V3 staking-vault case study.
- **Adaptive Tokenomics.** Industrial control engineering (PID, Proportional-Integral-Derivative) applied to token-supply emission as programmable monetary-policy mechanism; 806-run agent-based simulation battery; PID emission control cuts worst-case node-count variance from CV (coefficient of variation) 0.544 (static emission) to 0.087 (PID), functioning as downside insurance under competitor-shock attack.
- **Routing the Dollar.** Prepared for Federal Reserve / FRBNY Fifth Conference on the Dollar's International Roles (Jun 2026). Gateway concentration; CLII across 19 gateways with Tier 1 a 56% / 44% Coinbase-Circle duopoly; on Ethereum, transfer volume doubled while unique counterparties fell 25% and cross-gateway connections dropped from 5.8% to 2.6%.
- **Who Controls Agentic Payments?** First control-layer measurement of agentic payments (x402). May 2026 facilitator concentration: Base HHI 6,009 by count and 4,040 by dollar volume; Solana 4,203 and 4,915, all above the 2,500 highly-concentrated threshold at under 14 months of market age.
- **FSB AI Sound Practices response.** Filed 2026-07-01; the program's first international standard-setter filing. Three practice additions (composition-chain documentation, execution-authority evidence, dependency-concentration monitoring) plus the invited nonbank case study on settlement facilitators.
- **Minimum Viable Equivalence Pack.** Ten-category institutional diligence standard; finds bank-chartered stablecoin issuers face ~22x the regulatory capital requirements of money-transmitter-chartered issuers at $40B scale; production-validated by Circle's BUIDL March 27, 2025 incident (23 hours of limited availability). Editor-invited submission to the Journal of Financial Regulation.
- **Three Tokenization Strategies.** Each of three strategies satisfies different MVEP categories with measurably different strength (Regulated Collateral Circuit, Multi-Issuer Yield Stack, Tokenized Equity Rails); Securitize as cross-strategy concentration risk.
- **Seven Dollars.** Cross-product framework for major dollar-denominated digital products; CLII / MVEP decoupling; stress propagation through dependency chains (SVB, Circle pool, Paxos).
- **Dollar v3.** Four digital-dollar objects (payment stablecoins, tokenized deposits, deposit tokens, yield wrappers); typology for inheritance of control environments, backstop expectations, and failure modes; load-bearing for institutional scoping of product shape and counterparty topology.
- **Auditing Governance Concentration Beyond Token Allocation.** Accepted for publication at Frontiers in Blockchain. Live-governance audit of 52 token protocols: launch allocation does not predict governance concentration; DePIN governance is more concentrated than DeFi after correction (Cohen's d = 0.65, a directionally robust medium effect).
- **Who Burns the Tokens?** Burn-mechanism design (carrier contracts vs. subscriptions) drives demand concentration: Helium's carrier-contract model concentrates ~90% of token burns in 5 purchasers (HHI 0.22) despite ~1M hotspots; coalition-power analysis extends standard concentration measures for single-address risk.

TECHNICAL

Methods: policy-shock impact estimation (difference-in-differences), Monte Carlo simulation, PID controller design (industrial control engineering), coalition-power analysis, supply equilibrium modeling, build-versus-partner framework design

Data: Dune Analytics, Nansen Pro, Artemis, DefiLlama, CoinGecko, Helius (Solana), Etherscan, Blockscout (EVM), Spacescope, Token Terminal, Tally (DAO governance), rated.network, Beaconcha.in (validator data), FRED (macro); automated SEC filing and press release monitoring

Stack: Python (pandas, statsmodels, scipy, matplotlib), R (fixest, did, synthdid), SQL (DuneSQL, Trino), JavaScript / TypeScript; 22 MCP and API connections wrapping the data sources above plus GitHub; Claude.ai + Claude Code + Cursor with 36-skill library and 17-file living-knowledge architecture; n8n, Apify, Git / GitHub

RECOGNITION

- Uniswap Foundation governance committee (2025)
- ETHDenver pitch judge (2025)
- Introduction to Tokenomics article publicly endorsed by Binance founder CZ on day of publication

FRAMEWORKS AUTHORED

- Control Layer Intensity Index (CLII), Minimum Viable Equivalence Pack (MVEP), MVEP-gated Conditional Pass-Through (MGCPT; AG19 Rec 7), Three-Category Reserve Disclosure (3CRD).

EDUCATION

High Point University | B.S.B.A., Business Administration (Magna Cum Laude) · Investment Club President · Lingnan University, Hong Kong (exchange) *2018*